**Practical 5: Modeling Data Flow Diagram & Control Flow Diagram**

**Aim: To model and understand the Data Flow Diagram (DFD) and Control Flow Diagram (CFD) for the Pharmacy Management Software Project.**

**Introduction: Data Flow Diagrams (DFD) and Control Flow Diagrams (CFD) are essential tools in system analysis and design. They help in visualizing the flow of data and control within a system, providing a clear overview of the system's functionality and processes.**

**Objective:**

- **To identify external entities and functionalities of the system.**
- **To model the flow of data across the system using DFD.**
- **To represent control flows within the system using CFD.**

**Theory:**

**Data Flow Diagram (DFD): A Data Flow Diagram provides a functional overview of a system. It bridges the gap between users, system analysts, and designers by illustrating how data moves through the system. DFDs start with a high-level context diagram and progressively detail the system through lower levels.**

**Graphical Notations for DFD:**

| Term | Notation | Remarks |
|---|---|---|
| External Entity | Rectangle (▭) | Represents sources or destinations of data. |
| Process | Circle (◯) | Represents processes or functions. |
| Data Store | Left-right open rectangle (▥) | Denotes where data is stored. |
| Data Flow | Directed arrow (→) | Indicates the flow of data. |

**Explanation of Symbols in DFD:**

- **Process: Represented by a circle (◯), indicating the system's functions. Names should reflect functionality.**
- **External Entity: Appears only in the context diagram. Represented by a rectangle (▭) to show interaction with external sources or destinations.**
- **Data Store: Shown as a left-right open rectangle (▥), representing repositories where data is stored.**
- **Data Flow: Depicted by directed arrows (→) showing the movement of data within or outside the system.**

**Context Diagram and Leveling DFD:**

- **Context Diagram (Level 0): Gives a broad overview of the system as a single process with interactions from external entities.**

- **Leveling DFD: Splits the context diagram into sub-processes for detailed representation. Each level explores more granular system details.**
- **Balancing DFD: Ensures consistency between input/output data across levels. Data inflows and outflows must match in all subsequent levels.**

**Control Flow Diagram (CFD): A Control Flow Diagram illustrates the flow of control within a system, indicating the order of operations and control dependencies. It is useful for understanding decision points, conditional paths, and the sequence of actions.**

**Graphical Notations for CFD:**

| **Term** | **Notation** | **Remarks** |
|---|---|---|
| **Process/Action** | **Rectangle (▭)** | **Represents actions or processes in the system.** |
| **Decision** | **Diamond (◇)** | **Indicates decision points with multiple outcomes.** |
| **Control Flow** | **Directed arrow (→)** | **Shows the sequential flow of control.** |

**Case Study: Pharmacy Management System The Pharmacy Management System handles operations such as inventory management, billing, and sales tracking. The DFD and CFD for this system represent the flow of data and control as follows:**

**DFD (Level 0) - Context Diagram:**

- **External Entities: Customers, Suppliers**
- **Processes: Pharmacy Management System (Main Process)**
- **Data Stores: Inventory Database, Sales Records**
- **Data Flows:**
    - **Customer orders flow to the system.**
    - **Inventory details flow from the database to the system.**
    - **Sales records are updated in the database.**

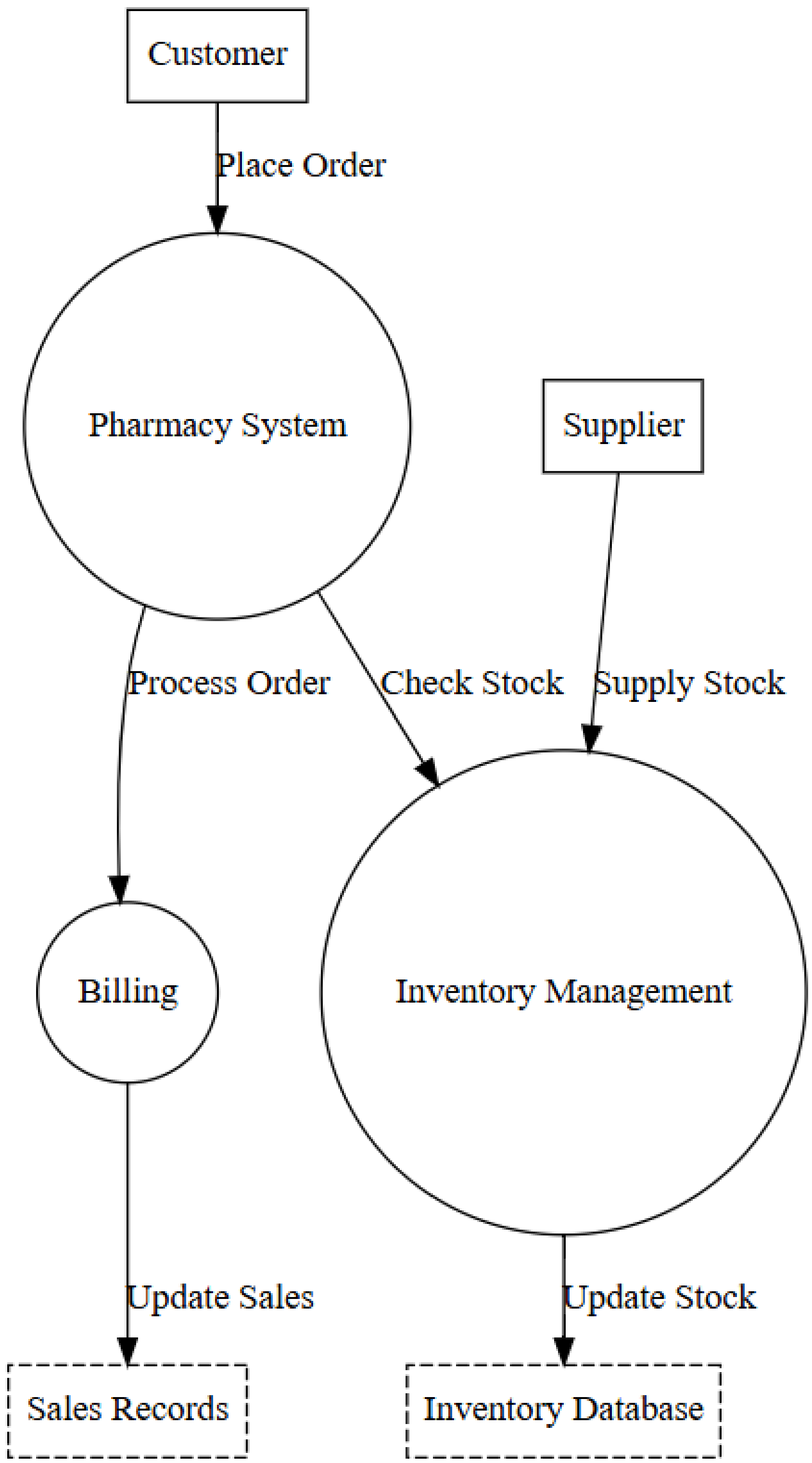
**DFD (Level 1):**

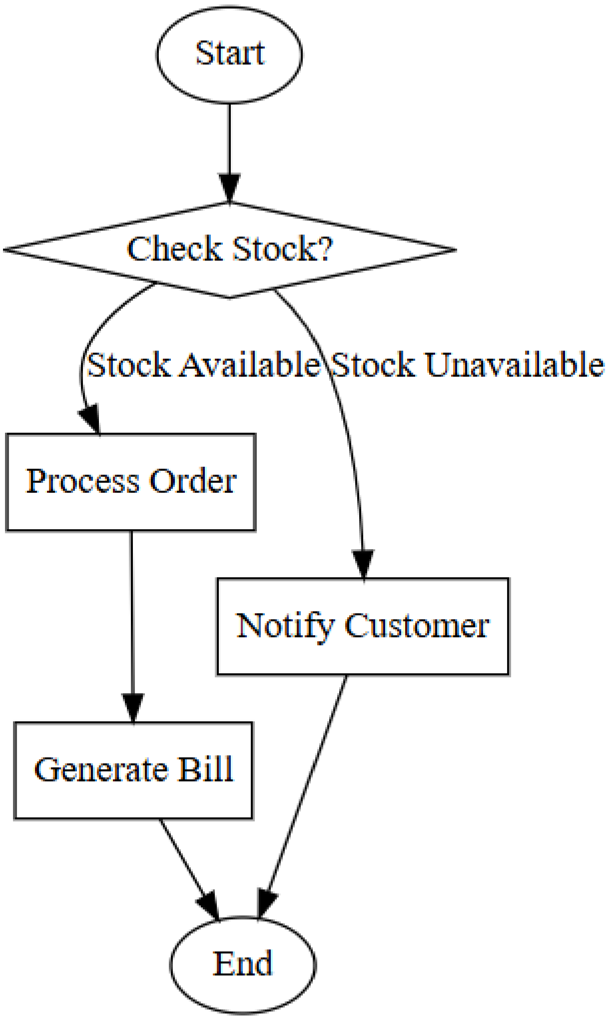
- **Processes:**
    - **Order Processing**
    - **Inventory Management**
    - **Billing**
- **Data Stores:**
    - **Order Records**
    - **Inventory Database**

**CFD Example:**

- **Process: Sales Processing**
- **Decision: Check Stock Availability (◇)**
- **Actions:**
    - **If stock is available, approve the sale and generate the bill.**
    - **If stock is unavailable, notify the customer.**
- **Control Flow:**
    - **Starts from customer order → Check stock → Generate bill/Notify customer**

**Conclusion: Modeling DFDs and CFDs for the Pharmacy Management Software Project provides a clear understanding of data movement and control mechanisms. These diagrams facilitate effective communication among stakeholders and aid in system design and implementation.**


Customer
Place Order
Pharmacy System
Supplier
Process Order
Check Stock
Supply Stock
Billing
Inventory Management
Update Sales
Update Stock
Sales Records
Inventory Database


Start
Check Stock?
Stock Available
Stock Unavailable
Process Order
Notify Customer
Generate Bill
End